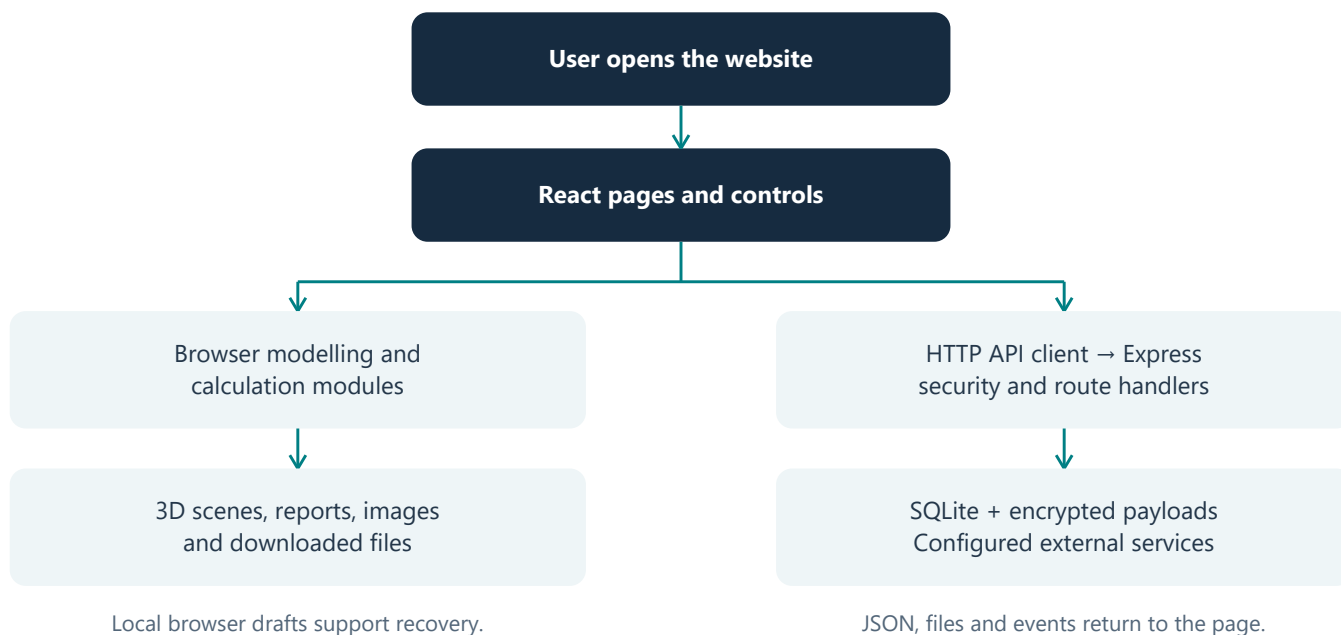


Secure-Nexus Website working workflow

How the website starts, processes requests and produces results

Source snapshot: **upgrade/r1-modeling-core** | 3 October 2026

The website consists of a **React browser application**, a **Node.js / Express API server** and a **SQLite database**. Most modelling, calculations and rendering happen in the browser. The server handles accounts, permissions, saving, collaboration and external services.



Startup sequence

- 1. Server:** configuration loads, SQLite opens, schema setup runs, and Express registers security middleware and API routes.
- 2. Website delivery:** production Express serves the compiled React website and /api endpoints from the same origin. Development Vite uses port 5173 and proxies API requests to port 4000.
- 3. Browser:** main.tsx loads routing, authentication, theme and notifications.
- 4. Session discovery:** the authentication provider calls /api/auth/bootstrap and /api/auth/me. Routing displays public pages or the authenticated dashboard.

This document describes inspected source. The website, tests, builds and native tools were not executed for this review.

Browser design and analysis modules

Most visible changes are produced locally. Editing updates browser-held state, the scene is refreshed, and saving is a separate API operation.

Feature / processor	Input → processing → output
Interior design Editor + Canvas3D	Room dimensions, furniture, colours and photos → browser state and Three.js scene → interactive room view and exportable model.
Residential / commercial CommunityEditor + CommunityScene	Buildings, levels, grids, openings and amenities → parametric geometry, structural screening, MEP planning and takeoff → community model, warnings and quantities.
Infrastructure InfraEditor + InfraScene	Highway, airport, port or dam parameters → facility geometry, terrain/civil planning and quantities → infrastructure scene and planning reports.
Engineering EngineeringWorkbench	SI input JSON or converted project data → frame/network analysis libraries → calculation reports and solver input decks.
BIM / survey / civil ExchangeWorkbench	IFC, IDS, survey files and alignment inputs → parsing, validation, supported geometry, terrain and civil calculations → exchange files, surfaces and reports.
Geometry / rendering GeometryWorkbench	Scene JSON or streaming manifest → viewport and background worker → clash findings, rendered PNG and benchmark results.
Documentation / export Export controls and browser libraries	Current design, views, schedules and quantities → document/file generation → PDF, DXF, IFC, BCF, OBJ, GLB, CSV, JSON or PNG, depending on the export path.

Editor selection

The project type selects the guided editor: interiors use the room workflow; residential, villa-community, townhouse and commercial projects use CommunityEditor; highways, airports, ports and dams use InfraEditor. The parent Editor manages project loading, save state and recovery.

Engineering processing and outputs

Inside the engineering workbench, the selected module determines which calculation handles the input.

Selected module	Processor / result
Structural frame	Frame analysis library → 2D displacements, reactions, member forces, convergence and supplied design-criteria checks.
Water / HVAC	Fluid network library → flows, heads or pressures, velocities, residuals and convergence.
Electrical	Circuit analysis → demand current, voltage drop, fault currents and protection checks.
Fire flow	Fire-flow assessment → terminal flow/pressure and storage-duration checks.
Equipment	Pump/fan selection → duty-point and capacity checks using supplied equipment curves.

Where computation runs

Engineering and exchange calculations currently run in the **browser UI thread**. Geometry clashes, path tracing and synthetic benchmarks run in a **Web Worker**, which sends progress and results back to the page. Structural coordination review also has a background worker.

Three output channels

- **Screen output:** scenes, reports, warnings, progress and saved status.
- **Downloads:** browser-generated documents/models/images or server-stored artifacts.
- **Stored output:** encrypted server records, project/workspace history and local recovery drafts.

The server workspace API stores calculation inputs and outputs; it does not execute engineering solvers or rendering jobs. Reports remain within their implemented mathematical and geometry subsets.

API modules and server responses

Request path: page → API client → Express middleware → route handler → permission/input checks → storage or external service → response → page update.

Middleware applies security headers, request limits and CSRF checks. Protected handlers also check the session and relevant project or organization permissions.

Request family	Responsible module / output
/api/auth	Authentication, security, email and TOTP handlers → registration/verification results, login session, user profile, session management and 2FA results.
/api/projects	Project routes, access checks, encryption and database → project lists, decrypted designs, save acknowledgments, photos, named snapshots and share-link management.
/api/projects/:id/workspaces	Workspace routes and storage → independent engineering/exchange/geometry revisions, encrypted artifacts, history and downloads.
/api/collaboration	Collaboration, access and lock modules → comments/issues, membership, presence, locks and server-sent events.
/api/organizations /api/sso	Organization and OIDC handlers → organization roles/project access, audit records and configured identity-provider login.
/api/payments	Pricing and payment-provider handlers → plans, payment records, checkout links and confirmed account-plan updates.
/api/assistant/plan	Assistant handler → validated action plan returned for client preview and explicit application.
/api/share/:token	Public share handler → authorized read-only project metadata and design JSON.
/api/audit	Audit routes → user event history, filters and statistics.
/api/secrets	Secret-vault handler → owner-authorized encrypted-secret storage and retrieval; no corresponding client page currently exists.
/api/cad-exchange/status	CAD provider registry → provider availability/status information. This endpoint does not currently perform DWG conversion.

Saving, revisions and recovery

Example: move an object and save the design

1. The editor changes its browser-held design.
2. Geometry and the visible scene update locally.
3. A browser recovery draft records the pending change.
4. After approximately **1.2 seconds**, the editor sends a project PATCH containing the design and its last known revision.
5. The server checks permissions, revision and applicable locks, encrypts the design, and writes it to SQLite.
6. The response returns the new revision; the editor updates its saved status.
7. Collaboration notifications can tell other users that a newer design exists.

If another user saved first, the server returns a revision conflict. The editor retains local work for retry or explicit comparison/merge. Normal editor saving is **revision-based**, rather than an automatic geometry CRDT.

Calculate and save a workbench report

1. Select a project and load or import inputs.
2. Calculate locally in the relevant browser module.
3. Review the report.
4. Choose **Save as new revision**.
5. The workspace panel uploads the payload and attached sources, then submits the workspace revision.
6. The server checks permissions and revision consistency and stores encrypted artifacts.

Workspace saves preserve CAD geometry and its revision. Original source binaries must be attached explicitly when exact source retention is needed. Engineering and Geometry retain reports; Exchange reconstructs calculated outputs from saved inputs during reload.

Current boundaries and source reference

Implemented source paths and pending integrations are separate. These boundaries affect how the website currently processes requests.

- **DWG conversion and native solver execution remain pending.** Existing solver adapters export input decks; they do not run OpenSees, STAAD or EPANET.
- **AI requires provider configuration.** Without a key, it returns a configuration message and empty plan. Applying supported actions uses the normal editor save flow.
- **Public sharing displays JSON**, rather than a complete public 3D viewer.
- **Payments default to demo mode.** Razorpay has webhook completion; the live PayPal client completion path is incomplete.
- **Google Maps loading conflicts with the current server CSP**, so that integration needs correction before being treated as operational.
- **Browser recovery is available**, but a complete offline/PWA website is still pending.

Verification status

This is the implemented processing map, with runtime verification deferred under the no-run instruction. No website execution, tests, builds or native converter/solver execution was performed for this workflow review. PDF generation and layout checks are separate document-authoring operations.

Source snapshot

Repository: [jitinraavi/Secure-Nexus](#)

Branch: [upgrade/r1-modeling-core](#)

Commit: [bd72532f2991c46922f2e40e504d1d2d42713b7d](#)

Browser entry and routing - [client/src/main.tsx](#)

API server entry - [server/src/index.ts](#)

Editor loading, saves and recovery - [client/src/pages/Editor.tsx](#)

Engineering workbench - [client/src/pages/EngineeringWorkbench.tsx](#)

BIM / survey / civil exchange - [client/src/pages/ExchangeWorkbench.tsx](#)

Geometry and rendering - [client/src/pages/GeometryWorkbench.tsx](#)

Workspace storage and history - [server/src/workspaces.ts](#)